Available online at www.sciencedirect.com
ScienceDirect
Information Processing Letters 101 (2007) 203–208
Information Processing Letters
www.elsevier.com/locate/ipl

A linear time algorithm for max-min length triangulation of a convex polygon

Shiyan Hu

Department of Computer and Information Science, Polytechnic University, Brooklyn, NY 11201, USA

Received 24 October 2005; received in revised form 15 September 2006; accepted 27 September 2006

Available online 17 November 2006

Communicated by F.Y.L. Chin

Abstract

We consider the following *planar max-min length triangulation* problem: given a set of n points in the Euclidean plane, find a triangulation such that the length of the shortest edge in the triangulation is *maximized*. In this paper, a linear time algorithm is proposed for computing the max-min length triangulation of a set of points in convex position. In addition, an $O(n \log n)$ time algorithm is proposed for computing the max-min length k-set triangulation of a set of points in convex position, where we are to compute a set of k vertices such that the max-min length triangulation on them is minimized over all possible k-set. We further show that the graph version of max-min length triangulation is NP-complete, and some common heuristics such as greedy algorithm are in general not able to give a bounded-ratio approximation to the max-min length triangulation.

Keywords: Computational geometry; Max-min length triangulation; Max-min length k-set triangulation

1. Introduction

A wide variety of triangulations targeted for different optimality have been studied. Among them, three Euclidean length-based triangulations, namely, minimum weight triangulation, maximum weight triangulation and min-max length triangulation, have received special research attention. Formally, let P be a set of n points in the plane. A *triangulation* $T(P)$ of P is a maximal set of non-intersecting straight-line segments connecting points in P. The *weight* $|T(P)|$ of $T(P)$ is the sum of the lengths of edges in $T(P)$. The *minimum weight triangulation* is a triangulation of P with minimum weight. The *maximum weight triangulation* is a triangulation of P with maximum weight. The *min-max length triangulation* is a triangulation of P such that the length of the longest edge is minimized. The *max-min length triangulation* is a triangulation of P such that the length of the shortest edge is maximized.

Minimum weight triangulation is a well known problem in computational geometry. Its complexity status remains unknown for a long time until recently it is shown to be NP-hard by Mulzer and Rote [13]. When restricted to a convex polygon, the minimum weight triangulation can be computed in $O(n^3)$ time by dynamic programming [7]. For general cases, one research direction is to compute the approximate solution. In [10], Levcopoulos and Krznaric propose an algorithm which produces a (very large) constant factor approximation to the optimal solution in $O(n \log n)$ time. Recently, a quasi-

E-mail address: shu@cis.poly.edu (S. Hu).

0020-0190/$ – see front matter © 2006 Published by Elsevier B.V.
doi:10.1016/j.ipl.2006.09.014

polynomial time approximation scheme is designed by Remy and Steger in [15]. Another research direction is to identify a subgraph of the minimum weight triangulation. Major results in this direction include *inclusion region* method and LMT-skeleton method. A well-known result on inclusion region method explores β-skeleton, where an edge e is included in the β-skeleton if two disks of diameter $\beta|e|$ passing through both endpoints of e contain no further point. Cheng and Xu [4] show that 1.1768 β-skeleton belongs to minimum weight triangulation, which is just slightly larger than the lower bound of 1.1603 for β shown by Wang and Yang [17]. $\mathrm{LMT}(P)$ is the set of edges which belong to every locally minimum triangulation of P, and $\mathrm{LMT}(P)$ is clearly a subgraph of minimum weight triangulation. LMT-skeleton is a large subgraph of $\mathrm{LMT}(P)$ which can be easily identified. It is found by Bose et al. [2] that $\mathrm{LMT}(P)$ generates $\Theta(n)$ components with probability approaching one for the point set randomly generated according to uniform distribution.

As a natural complementary problem, maximum weight triangulation has also received research attention recently. The first work on it appears in [16], where an $O(n^2)$ algorithm is presented to find the maximum weight triangulation of an n-sided polygon inscribed in a disk. Later, the author [8] proposes the spoke triangulation which leads to a 6-approximation for maximum weight triangulation of the point set in general position. The approximation ratio is improved to 4.238 in [5] through giving better weight bound on maximum weight triangulation. When restricted to a convex polygon, a linear-time approximation scheme exists [14]. The third length-based triangulation problem which receives investigation is min-max length triangulation. The optimal solution can be computed in $O(n^2)$ time [6].

In contrast to the above results, there is no previous work on computing max-min length triangulation which refers to the triangulation where the length of the shortest edge is maximized. Theoretically, max-min length triangulation is as important as the other three length-based triangulations, and should attract sufficient research attention in order to better understand the nature of optimal triangulations. It is interesting to note that some triangulation problems based on other "max-min" metrics have been studied. Some recent advances include max-min height triangulation [1], max-min height covering triangulation [18], and max-min area triangulation [9].

In this paper, we investigate the max-min Euclidean length triangulation problem and propose a linear time algorithm for computing the max-min length triangulation of a set of points in convex position. In addition, an $O(n \log n)$ time algorithm is proposed for computing the max-min length k-set triangulation of a set of points in convex position, where the problem is to compute a set of k vertices such that the max-min length triangulation on them is minimized over all possible k-set. We further show that the graph version of max-min length triangulation is NP-complete and common heuristics such as greedy algorithm and maximum non-crossing spanning tree heuristic are not even able to give a bounded-ratio approximation to the max-min length triangulation.

The rest of the paper is organized as follows: Section 2 describes the max-min length triangulation algorithm as well as the max-min length k-set triangulation algorithm. Section 3 presents some "hardness" results on the max-min length triangulation problem. A summary of the work is given in Section 4.

2. The algorithms

2.1. Computing max-min length triangulation for a convex polygon

We are to present a linear time algorithm for computing max-min length triangulation of a convex polygon. We first note the following fact.

Observation 1. *In any convex polygon, there are at most two inner angles which can be smaller than $\frac{\pi}{3}$.*

Proof. It is well known that the sum of total inner angles equals $(n-2)\pi$ for any convex polygon. Suppose to the contrary, we have three inner angles smaller than $\frac{\pi}{3}$. The sum of inner angles of the convex polygon must then be strictly less than $(n-3)\pi + 3 \times \frac{\pi}{3} = (n-2)\pi$ which leads to contradiction. □

In the algorithm, we begin with any vertex p and assume that q, r are the next two vertices on the convex polygon in clockwise order. We first check q. If the inner angle of q (i.e., $\angle pqr$) is no smaller than $\frac{\pi}{3}$, we "cut" the corner q through linking pr and continue to process r. It is critical to note that since $\angle pqr \geqslant \frac{\pi}{3}$, pr is no shorter than at least one of pq and qr in $\triangle pqr$. In other words, the inserted edge pr is no shorter than the shortest edge on the convex polygon. It is possible that $\angle pqr < \frac{\pi}{3}$, we then proceed to the next hull vertex r. If the inner angle of r is also smaller than $\frac{\pi}{3}$, its next vertex must have inner angle at least $\frac{\pi}{3}$ by Observation 1.

At a high level, we process each vertex and its inner angle in a clockwise order. A vertex can be either cut or left untouched depending on the comparison result of its inner angle with $\frac{\pi}{3}$. The algorithm runs in linear time since we at most need to process four vertices to find a vertex to be cut (i.e., to add an edge), and in a total, we only need to add a linear number of edges for the triangulation. The algorithm is clearly optimal in terms of running time. Since each time we always add an edge no shorter than some previously added edges (or hull edges), another straightforward observation is that a shortest edge of any max-min length triangulation for a convex polygon is a shortest edge on the convex polygon. We reach the following theorem:

Theorem 2. *Given a convex polygon, the max-min length triangulation on hull vertices can be computed in linear time and a shortest edge in the triangulation lies on the convex polygon.*

Refer to Fig. 1 for an example of carrying out our algorithm. Suppose that we start from a and thus first check b. Since $\angle abc > \frac{\pi}{3}$, we cut b and proceed to c. We then check d and cut it since $\angle cde > \frac{\pi}{3}$. We reach e and check f, we leave it untouched since $\angle efa < \frac{\pi}{3}$. We then check a and cut it since $\angle fac > \frac{\pi}{3}$. The algorithm terminates because enough edges have been inserted.

2.2. Computing max-min length k-set triangulation for a convex polygon

We next investigate the max-min length k-set triangulation problem. The class of k-set problems, which are characterized by picking "best" k vertices out of the whole point set according to some optimality criteria (such as k-MST problem [12]), have received much research attention in computational geometry. In our context, the problem is defined as follows: compute a set of k ($k \leqslant n$) vertices (called a k-set) such that the max-min length triangulation on these vertices is minimized over all possible k-set. Consider that performance of interconnected network in some sense degrades as "distance" between the network nodes increases. The proposed algorithm can be used to compute a subset of network nodes such that even the worst-case interconnection performance, which is measured with the shortest edge in any possible triangulation on them, is still acceptable.

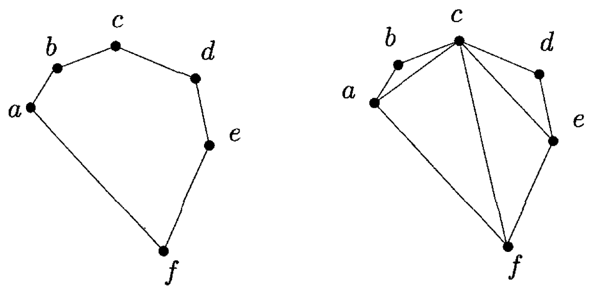

Fig. 1. An example of running the max-min length triangulation algorithm on a convex polygon.

By Theorem 2, we only need to compute a set of k vertices such that the shortest hull edge on them is minimized over all possible k-set. It is easy to come up with an $O(n^3)$ algorithm for this purpose by dynamic programming. A faster algorithm is possible by exploring some specific prosperities of the problem.

The main observation is that at most n inner edges (i.e., edges not on the convex hull) can be shorter than the shortest hull edge which is denoted by h_{short}.[1] Our algorithm is as follows. We first compute the $n+1$ shortest edges, and then test for each edge whether there are at least k points to one side of it. During the process, we record the shortest edge, denoted by "min edge", which passes the test. Note that an h_{short} must be among these $n+1$ edges and thus at least one edge will pass the test. To a side of "min edge", there are at least k vertices. We then arbitrarily select k vertices there and carry out the max-min length triangulation algorithm on them. The returned triangulation is an optimal max-min length k-set triangulation since the shortest edge in the resulting triangulation is exactly the "min edge" .

Our main task becomes to prove that at most n inner edges can be shorter than h_{short}. For this, it is sufficient to prove that for an inner edge e shorter than h_{short}, any inner edge properly crossing e must be longer than h_{short}. We begin with the following fact:

Observation 3. *For any triangle $\triangle pqr$ with base edge pq, if pq is shorter than all edges along a convex path inside the triangle (starting from p and ending with q), the bounding triangle for the convex path and pq must have vertex angle (opposite to pq) smaller than $\frac{\pi}{3}$.*

For Fig. 2, Observation 3 says that the angle $\angle pcq$ formed by the line extending pa and the line extending qb is smaller than $\frac{\pi}{3}$ (since $|pq| < |pa| \leqslant |pc|$ and $|pq| < |qb| \leqslant |qc|$). It further follows that $\angle prq < \frac{\pi}{3}$.

Refer to Fig. 3. Suppose that two crossing inner edges pq and or are both shorter than h_{short}, and without loss of generality, assume that $|pq| \geqslant |or|$. It is easy to see that at least one of po, qr and at least one of pr, qo must be shorter than pq by triangle inequality.

[1] More than one hull edges can have length $|h_{\text{short}}|$.

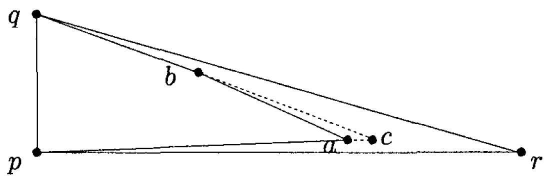

Fig. 2. A convex path within a triangle.

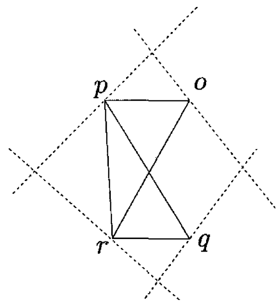

Fig. 3. No three edges can be shorter than pq.

We claim that actually at most two of po, oq, qr, pr can be shorter than pq. Otherwise, without loss of generality, we assume that po, pr, qr are shorter than pq and thus h_{short}. Therefore, none of them are hull edges and there must have certain hull vertices above them. Subsequently, hull edges above these three edges are all longer than them. Let dotted lines in Fig. 3 be those "bounding" lines such that convex polygon lies to only one side of each line. By Observation 3, it follows that three out of four angles (namely, the angles facing po, pr and qr) formed by dotted lines are smaller than $\frac{\pi}{3}$. This contradicts the fact that any convex quadrilateral can have at most two inner angle smaller than $\frac{\pi}{3}$.

Therefore, we can have at most two edges out of po, oq, qr, pr shorter than pq. Without loss of generality, we assume that only po and pr are shorter than pq and thus h_{short}. Since $|ro| \leqslant |pq| \leqslant |oq|, |qr|$, $\angle rqo \leqslant \frac{\pi}{3}$ in $\triangle rqo$. Note that there must have vertices above po and pr since they cannot be hull edges. We are to focus on the region shown in Fig. 4, which is formed by extending qo to intersect l at u, and extending qr to intersect l at v. From the above discussion, one easily sees that $\angle pvr < \frac{\pi}{3}$ and $\angle puo < \frac{\pi}{3}$. Since $\angle rqo \leqslant \frac{\pi}{3}$, we contradict the fact that qvu forms a triangle.

We have shown that no pair of crossing inner edges can be both shorter than h_{short}. It readily follows that one can have at most n edges shorter than h_{short} in any convex polygon. A convex polygon with $\frac{n}{2} - 1$ edges shorter than h_{short} is shown in Fig. 5.

We are ready to prove the following theorem:

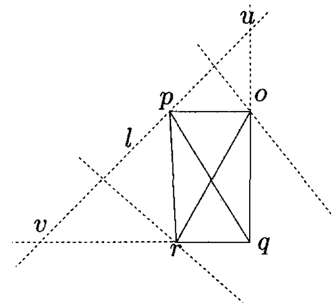

Fig. 4. Crossing edges cannot be both shorter than h_{short}.

Fig. 5. $\frac{n}{2} - 1$ edges shorter than h_{short}.

Theorem 4. *Given a convex polygon, the max-min length k-set triangulation on n hull vertices can be computed in* $O(n \log n)$ *time.*

Proof. Since at most n inner edges can be shorter than h_{short}, we can first compute the $n + 1$ shortest edges for the point set, and then test for each edge whether there are at least k points to one side of it. The shortest edge which passes the test is the "min edge". We then arbitrarily select k points to one side of it and the max-min length triangulation on them is an optimal max-min length k-set triangulation since the "min edge" is the shortest edge in the triangulation.

Identifying "min edge" involves edge enumeration and edge test. Enumerating the shortest $n + 1$ edges among all $\Theta(n^2)$ edges on n hull vertices can be performed in worst-case $O(n \log n + n + 1) = O(n \log n)$ time [3]. To decide whether there are at least k points to one side of an edge, we sort the hull vertices in the clockwise order (as a pre-processing step) and record the rank for each vertex. The number of points to one side of an edge can then be easily computed in $O(1)$ time. After "min edge" is identified, the max-min length triangulation can be computed in $O(k)$ time. It is clear that the running time of the whole algorithm is bounded above by that of edge enumeration, which is $O(n \log n)$ time. □

3. Hardness results

We first show that the graph version of the max-min length triangulation problem is NP-complete. We reduce from the well-known *triangulation existence problem* (TE): given a planar point set P and a set E of edges, decide whether E contains a triangulation of P.

Theorem 5. (Lloyd [11]) *TE is* NP*-complete.*

Max-Min length Triangulation for undirected planar Graph problem (MMTG) can be described as follows. Given a complete graph $G = (P, E)$, positive rational weight (or length) $w(e)$ for each edge $e \in E$, and a positive number k, decide whether there is a triangulation $T(P)$ in G such that $\min_{e \in T(P)} w(e) \geqslant k$.

Theorem 6. *MMTG is* NP*-complete.*

Proof. It is easy to verify that MMTG belongs to NP. We reduce from TE to show that MMTG is NP-hard. Let $G = (P, E)$ where $|P| = n$ be an instance of TE. Let $G' = (P, E')$ be the complete graph of P. We assign weight to each edge in E' as follows:

$$w(e') = \begin{cases} 1: & e' \in E, \\ 0: & e' \notin E. \end{cases}$$

We claim that there is a triangulation in G if and only if there is a triangulation in G' with the shortest edge of length at least $k = 1$. If G contains a triangulation, then clearly G' contains a triangulation with the shortest edge length of k. On the other hand, if G' contains a triangulation with the shortest edge length of at least k, then no zero-length edge can be included in the triangulation. It follows that G contains a triangulation. $\square$

We next show that some common heuristics cannot produce good approximation to max-min length triangulation for a general point set in Euclidean plane.

First, edge-flipping strategy cannot produce good solutions in general. In our case, edge flipping strategy begins with an arbitrary triangulation and successively refines it until every convex quadrilateral is optimally triangulated. Here, a convex quadrilateral is optimally triangulated when the longer diagonal is inserted. Essentially, we are to show that locally optimal triangulation might not be globally optimal in some cases. Refer to Fig. 6 for such an example. In Fig. 6, b and c are very close to each other (i.e., $|bc|$ is an infinitesimal value) while other points are well apart subject to the constraints that $|fa|, |fd|, |fb|, |fc| > |ad|, |cd|$. The solid lines form a locally optimal triangulation which can be returned by edge-flipping strategy and the dotted lines lead to the globally optimal triangulation. Clearly, the approximation ratio of the edge-flipping algorithm is even *unbounded* in this case (precisely, the approximation ratio is $|ac|/|bc|$ where $|bc|$ is an infinitesimal value and ac is the shortest edge in the max-min length triangulation as shown in Fig. 6).

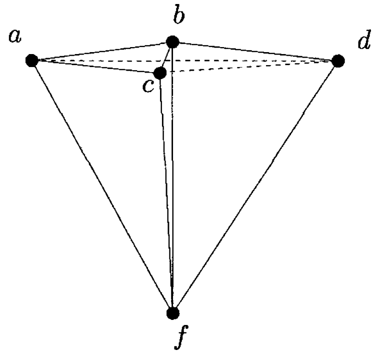

Fig. 6. A counterexample for carrying out common heuristics.

Similar argument goes to greedy algorithm. In our context, greedy algorithm is to repeatedly add a longest possible edge that does not properly intersect any of the previously generated edges. Therefore, in Fig. 6, fa, fd, fb, fc are first inserted and thus greedy algorithm cannot produce a good solution. Maximum non-intersecting spanning tree heuristic does not work either. In this heuristic, the maximum non-intersecting spanning tree is constructed on the point set followed by completing triangulation by any other method. Since fa, fd, fb, fc are included in the maximum non-intersecting spanning tree, we can only obtain the same result as before. In summary, none of these common heuristics are applicable for approximating the max-min length triangulation.

4. Conclusion

This paper is the first report of studying max-min length triangulation of a point set in Euclidean plane. The work may help us better understand the nature of optimal triangulations. The contribution of this paper includes both positive results and negative results. The positive results include an optimal linear time algorithm for computing the max-min length triangulation of a convex polygon. In addition, the max-min length k-set triangulation of a convex polygon can be computed in $O(n \log n)$ time. The negative results include that the graph version of the max-min length triangulation problem is NP-complete and several common heuristics are not applicable for computing the general max-min Euclidean length triangulation. From our study, it seems hard to come up with a *bounded-ratio* approximation algorithm for triangulating a general point set with max-min length as optimality criterion.

References

[1] M.W. Bern, H. Edelsbrunner, D. Eppstein, S.L. Mitchell, T.S. Tan, Edge insertion for optimal triangulations, Discrete & Computational Geometry 10 (1993) 47–65.

[2] P. Bose, L. Devroye, W. Evans, Diamonds are not a minimum weight triangulation's best friend, International Journal of Computational Geometry 12 (6) (2002) 445–453.
[3] T.M. Chan, On enumerating and selecting distances, International Journal of Computational Geometry and Applications 11 (2001) 291–304.
[4] S.-W. Cheng, Y.-F. Xu, On β-skeleton as a subgraph of a minimum weight triangulation, Theoretical Computer Science 262 (2001) 459–471.
[5] F. Chin, J. Qian, C. Wang, Progress on maximum weight triangulation of a point set, in: Proceedings of the International Conference on Computing and Combinatorics (COCOON), in: Lecture Notes in Computer Science, vol. 3106, Springer, Berlin, 2004, pp. 53–61.
[6] H. Edelsbrunner, T.S. Tan, A quadratic time algorithm for the min-max length triangulation, SIAM Journal on Computing 22 (3) (1993) 527–551.
[7] P.D. Gilbert, New results in planar triangulations, Master's thesis, University of Illinois, Urbana, 1979.
[8] S. Hu, A constant approximation algorithm for maximum weight triangulation, in: Proceedings of the Canadian Conference on Computational Geometry (CCCG), 2003, pp. 150–154.
[9] J.M. Keil, T. Vassilev, Algorithms for optimal area triangulations of a convex polygon, Computational Geometry: Theory and Applications 35 (3) (2006) 173–187.
[10] C. Levcopoulos, D. Krznaric, Quasi-greedy triangulations approximating the minimum weight triangulation, Journal of Algorithms 27 (2) (1998) 303–338.
[11] E.L. Lloyd, On triangulations of a set of points in the plane, in: Proceedings of the IEEE Symposium on Foundations of Computer Science (FOCS), 1977, pp. 228–240.
[12] J.S.B. Mitchell, Guillotine subdivisions approximate polygonal subdivisions: A simple polynomial-time approximation scheme for geometric TSP, k-MST, and related problems, SIAM Journal on Computing 28 (4) (1999) 1298–1309.
[13] W. Mulzer, G. Rote, Minimum weight triangulation is NP-hard, in: Proceedings of ACM Symposium on Computational Geometry (SoCG), 2006, pp. 1–10.
[14] J.B. Qian, C.A. Wang, A linear-time approximation scheme for the maximum weight triangulations of a convex polygon, Algorithmica 40 (2004) 161–172.
[15] J. Remy, A. Steger, A quasi-polynomial time approximation scheme for minimum weight triangulation, in: Proceedings of ACM Symposium on Theory of Computing (STOC), 2006, pp. 316–325.
[16] C.A. Wang, F. Chin, B. Yang, Maximum weight triangulation and graph drawing, Information Processing Letters 70 (1) (1999) 17–22.
[17] C.A. Wang, B. Yang, A lower bound for beta-skeleton belonging to minimum weight triangulations, Computational Geometry: Theory and Applications 19 (1) (2001) 35–46.
[18] B. Zhu, X. Deng, On computing and drawing maxmin-height covering triangulation, in: Proceedings of the International Symposium on Graph Drawing (GD), in: Lecture Notes in Computer Science, vol. 1547, Springer, Berlin, 1998, pp. 464–466.